

Multiple Choice

1. **Answer:** A

Options: A) -7.89×10^3 J; B) -12.75×10^3 J; C) -8.01×10^3 J; D) -10.02×10^3 J; E) -11.32×10^3 J

Note: Correct option: A - -7.89×10^3 J

2. **Answer:** E

Options: A) Any craters that existed have been eroded through the strong winds on Io's surface.; B) Io is protected by a layer of gas that prevents any objects from reaching its surface.; C) The gravitational pull of nearby moons prevent objects from hitting Io.; D) Io's magnetic field deflects incoming asteroids, preventing impact craters.; E) Io did have impact craters but they have all been buried in lava flows.

Note: Correct option: E - Io did have impact craters but they have all been buried in lava flows.

3. **Answer:** A

Options: A) Mass, velocity, and acceleration; B) Mass, velocity, height, and acceleration; C) Mass and velocity; D) Velocity, height, and acceleration; E) Velocity and acceleration

Note: Correct option: A - Mass, velocity, and acceleration

4. **Answer:** E

Options: A) 1000 times more; B) 50 times more; C) 5000 times more; D) 500 times more; E) 10000 times more

Note: Correct option: E - 10000 times more

5. **Answer:** C

Options: A) reduces by 1; B) increases by 3; C) increases by 1; D) reduces by half; E) increases by 4

Note: Correct option: C - increases by 1

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'size of the planet'.

8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '6.75'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 10.0010^{-15} N. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 25,000 ergs. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key